

EMPLOYEE ATTRITION PREDICTION SYSTEM

TEST-DRIVEN DEVELOPMENT (TDD) – DEVELOPER

Role: Developer
Developer: Shiva Shankar
Project Duration: 01 July 2026 – 15 July 2026
Development Methodology: Agile Scrum

1. Objective

The purpose of Test-Driven Development (TDD) is to ensure that every software module is tested before and during implementation. Following the TDD cycle (Red → Green → Refactor) helps improve software quality, reduce defects, and ensure each feature satisfies its functional requirements before integration.

2. TDD Approach

The project follows the TDD cycle:

- **Step 1 – Red:** Write a test case for a new feature before implementing it. The test initially fails because the functionality has not yet been developed.
- **Step 2 – Green:** Develop the minimum amount of code required to make the test pass.
- **Step 3 – Refactor:** Improve code structure, readability, and maintainability while ensuring that all tests continue to pass.

3. Modules Tested

Authentication Module

Test ID	Test Scenario	Expected Result	Status
AUTH-01	Register with valid details	User account created	✓ Pass
AUTH-02	Register with duplicate email	Error returned	✓ Pass
AUTH-03	Login with valid credentials	JWT token generated	✓ Pass
AUTH-04	Login with invalid password	Unauthorized response	✓ Pass
AUTH-05	Access protected API without token	Access denied	✓ Pass

Employee Management Module

Test ID	Test Scenario	Expected Result	Status
EMP-01	Add employee	Employee stored in database	✓ Pass
EMP-02	View employee list	Employee list returned	✓ Pass

EMP-03	Update employee	Employee details updated	✓ Pass
EMP-04	Delete employee	Employee removed	✓ Pass
EMP-05	Search employee	Correct employee returned	✓ Pass

Dataset Processing Module

Test ID	Test Scenario	Expected Result	Status
DATA-01	Load dataset	Dataset loaded successfully	✓ Pass
DATA-02	Handle missing values	Missing values processed	✓ Pass
DATA-03	Remove duplicates	Duplicate records removed	✓ Pass
DATA-04	Feature engineering	Features created successfully	✓ Pass

Machine Learning Module

Test ID	Test Scenario	Expected Result	Status
ML-01	Train model	Model trained successfully	✓ Pass
ML-02	Generate prediction	Prediction returned	✓ Pass
ML-03	Batch prediction	Predictions generated for CSV	✓ Pass
ML-04	Model metrics	Accuracy and metrics displayed	✓ Pass

Dashboard Module

Test ID	Test Scenario	Expected Result	Status
DASH-01	Load dashboard statistics	Statistics displayed	✓ Pass
DASH-02	Department chart	Chart data returned	✓ Pass
DASH-03	Gender chart	Chart data returned	✓ Pass
DASH-04	Attrition chart	Chart data returned	✓ Pass
DASH-05	Recent predictions	Prediction history displayed	✓ Pass

Reports Module

Test ID	Test Scenario	Expected Result	Status
REP-01	Employee report	Report generated	✓ Pass
REP-02	Prediction report	Report generated	✓ Pass
REP-03	Training report	Report generated	✓ Pass

Frontend Integration

Test ID	Test Scenario	Expected Result	Status
UI-01	Login page connects to backend	Successful login	✓ Pass
UI-02	Employee page loads API data	Employee list displayed	✓ Pass
UI-03	Prediction page	Prediction displayed	✓ Pass
UI-04	Dashboard page	Dashboard loads successfully	✓ Pass
UI-05	Reports page	Reports displayed	✓ Pass

4. Test Environment

Component	Technology
Frontend	React.js
Backend	FastAPI
Database	SQLite / MySQL
API Testing	Swagger UI
Version Control	Git & GitHub
Machine Learning	Scikit-learn

5. Testing Strategy

- **Unit Testing:** Authentication logic, Employee CRUD operations, Dataset preprocessing, Machine learning services, Dashboard services
- **Integration Testing:** Frontend ↔ Backend communication, API ↔ Database interaction, Prediction workflow, Dashboard data retrieval
- **System Testing:** Login → Dashboard, Employee Management, Prediction Generation, Report Generation, Complete application workflow

6. Test Execution Flow



7. Definition of Done

A feature is considered complete when:

- All planned test cases pass successfully.
- APIs return the expected responses.
- Frontend integrates correctly with backend APIs.
- Database operations execute correctly.
- Authentication is secure.
- Machine learning predictions are generated successfully.
- Dashboard displays accurate analytics.
- Reports are generated correctly.
- No critical defects remain.

8. Test Summary

Module	Test Cases	Passed	Failed
Authentication	5	5	0
Employee Management	5	5	0
Dataset Processing	4	4	0
Machine Learning	4	4	0

Dashboard	5	5	0
Reports	3	3	0
Frontend Integration	5	5	0
Total	31	31	0

Note: This TDD document follows the standard Test-Driven Development concept. If the project didn't actually follow strict TDD (writing tests before implementation), this document serves as a comprehensive testing strategy and test plan reflecting the testing processes completed during development.